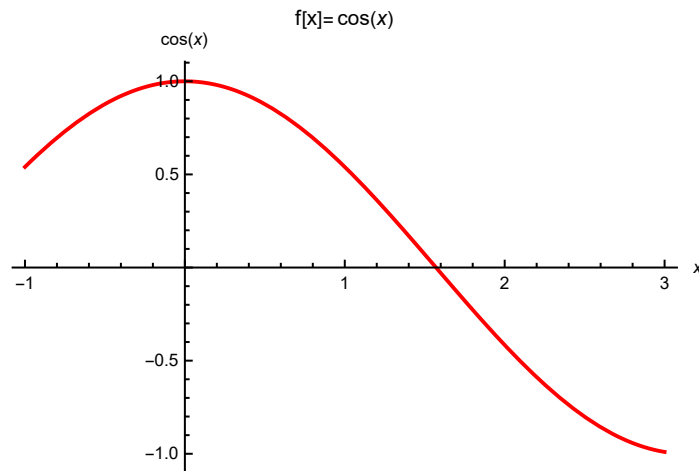


Practical 2(N-R Method)

Question1

```
In[ ]:= f[x_] = Cos[x];  
Plot[f[x], {x, -1, 3}, PlotStyle -> {Red, Thick},  
AxesLabel -> {x, f[x]}, PlotLabel -> "f[x] = " f[x]]
```

Out[]:=



```
In[ ]:= x0 = 1;  
Nmax = 20;  
eps = 10^(-6);  
Print["f[x]=", f[x]];  
Print["f[x]=", D[f[x], x]];  
For[i = 1, i ≤ Nmax, i++, x1 = N[x0 - (f[x] /. x -> x0) / (D[f[x], x] /. x -> x0)];  
If[Abs[x1 - x0] < eps, Return[x1], x0p = x0; x0 = x1];  
Print["In", i, "th Number of iteration the approximation to root is:", x1];  
Print["Estimated error is :", Abs[x1 - x0p]]];  
Print["the final approximation of root is:", x1];  
Print["Estimated error is :", Abs[x1 - x0]]
```

f[x]=Cos[x]

f[x]=-Sin[x]

In1th Number of iteration the approximation to root is:1.64209

Estimated error is :0.642093

In2th Number of iteration the approximation to root is:1.57068

Estimated error is :0.0714173

In3th Number of iteration the approximation to root is:1.5708

Estimated error is :0.00012105

Out[]:=

Return[1.5708]

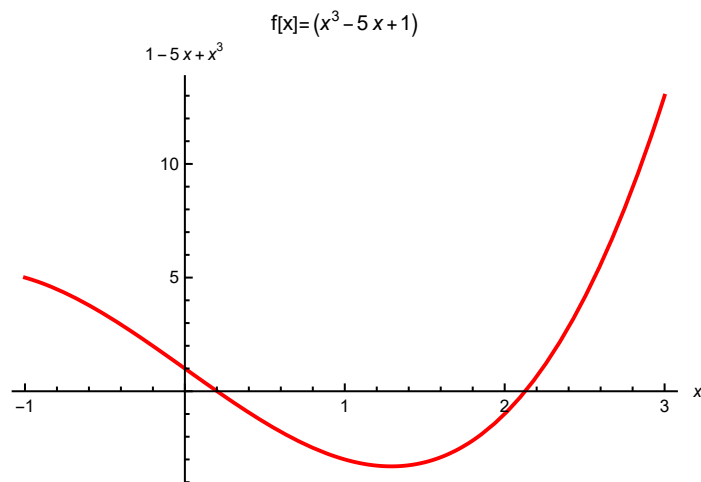
the final approximation of root is:1.5708

Estimated error is : 5.91305×10^{-13}

Question2

```
In[ ]:= f[x_] = x^3 - 5 * x + 1;
Plot[f[x], {x, -1, 3}, PlotStyle -> {Red, Thick},
AxesLabel -> {x, f[x]}, PlotLabel -> "f[x] = " f[x]]
```

Out[]:=



```
In[ ]:= x0 = 0.5;
Nmax = 20;
eps = 10^(-6);
Print["f[x] = ", f[x]];
Print["f[x] = ", D[f[x], x]];
For[i = 1, i <= Nmax, i++, x1 = N[x0 - (f[x] /. x -> x0) / (D[f[x], x] /. x -> x0)];
If[Abs[x1 - x0] < eps, Return[x1], x0p = x0; x0 = x1];
Print["In", i, "th Number of iteration the approximation to root is:", x1];
Print["Estimated error is :", Abs[x1 - x0p]];
Print["the final approximation of root is:", x1];
Print["Estimated error is :", Abs[x1 - x0]]
```

$f[x] = 1 - 5x + x^3$

$f[x] = -5 + 3x^2$

In1th Number of iteration the approximation to root is:0.176471

Estimated error is :0.323529

In2th Number of iteration the approximation to root is:0.201568

Estimated error is :0.0250975

In3th Number of iteration the approximation to root is:0.20164

Estimated error is :0.0000716007

Out[]:=

Return[0.20164]

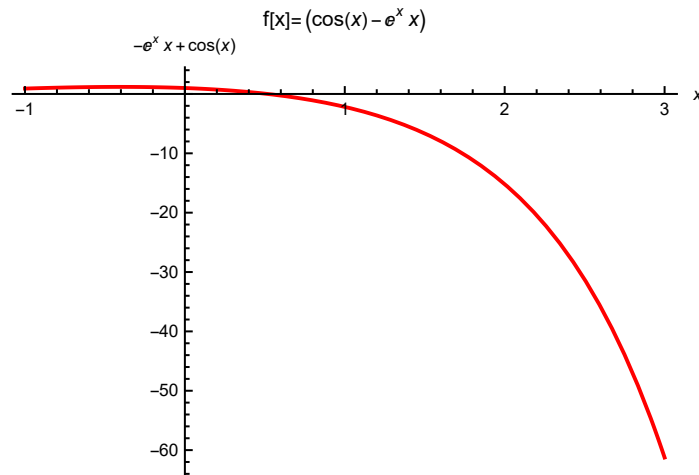
the final approximation of root is:0.20164

Estimated error is : 6.35602×10^{-10}

Question 3

```
In[*]:= f[x_] = Cos[x] - x * Exp[x];
Plot[f[x], {x, -1, 3}, PlotStyle -> {Red, Thick},
  AxesLabel -> {x, f[x]}, PlotLabel -> "f[x]=\" f[x]"]
```

Out[*]=



```
In[*]:= x0 = 1.5;
Nmax = 20;
eps = 10^(-6);
Print["f[x]=", f[x]];
Print["f[x]=", D[f[x], x]];
For[i = 1, i ≤ Nmax, i++, x1 = N[x0 - (f[x] /. x -> x0) / (D[f[x], x] /. x -> x0)];
  If[Abs[x1 - x0] < eps, Return[x1], x0p = x0; x0 = x1];
  Print["In", i, "th Number of iteration the approximation to root is:", x1];
  Print["Estimated error is :", Abs[x1 - x0p]]];
Print["the final approximation of root is:", x1];
Print["Estimated error is :", Abs[x1 - x0]]
```

$f[x] = -e^x x + \cos[x]$

$f[x] = -e^x - e^x x - \sin[x]$

In1th Number of iteration the approximation to root is:0.954848

Estimated error is :0.545152

In2th Number of iteration the approximation to root is:0.632019

Estimated error is :0.322829

In3th Number of iteration the approximation to root is:0.527616

Estimated error is :0.104403

In4th Number of iteration the approximation to root is:0.517838

Estimated error is :0.00977784

In5th Number of iteration the approximation to root is:0.517757

Estimated error is :0.0000806043

Out[*]=

Return[0.517757]

the final approximation of root is:0.517757

Estimated error is : 5.44033×10^{-9}